SIGma **Fall 2026, October 26th, 2026**

2027 CG:SHOP: lawn mowing approximation

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Last week, we discussed the traveling salesman problem (TSP), which (as in lawn mowing) seeks the shortest tour to cover some object. The distinction is that TSP runs on a combinatorial object, while lawn mowing is a geometric problem on the plane. My goal today is to illustrate the thought process behind generalization, through an instructive case study in [1]. While the specific results are not the main point, they will also be used in future weeks.

1 from lawn mowing to the TSP

We consider the lawn-mowing problem:

- **input**: a polygonal region R and a unit-square cutter χ
- **output**: an axis-aligned tour τ through points in $\mathcal{R}^2$ such that $\bigcup_{p\in\tau} \chi(p) \supseteq R$, minimizing $|\tau|$

The points of τ are continuous, so there are uncountably many tours to consider. As described in the introduction, an intuitive and reasonable approach is to find a *reduction* to traveling salesman, i.e. how can we use TSP can help us solve lawn mowing.

The input to TSP is a graph, namely a finite number of points, alongside a distance function (metric) between pairs of points. Looking at the lawn mowing problem, we need to find a way to discretize it. There are two obvious approaches:

- consider the graph based on all grid points in the plane
- consider the graph based on the "pixels", i.e. unit squares (cf. Fig 1), surrounding each grid point

We will take the second approach here, as we want to cover a given region of space (and so it is natural to consider a discretization as unit space). The next aspect is to determine which of these so-called pixels are relevant to our problem (as otherwise we have an infinite plane). We may classify pixels into three groups:

- pixels that are completely inside the region R
- pixels that overlap, but are not completely contained in R
- pixels that are completely outside the region R

Pixels completely outside R are not useful to us, and we drop them immediately. Pixels completely inside R should be included, because thats the area we want to cover. Then, there are the pixels that overlap; should we include or exclude?

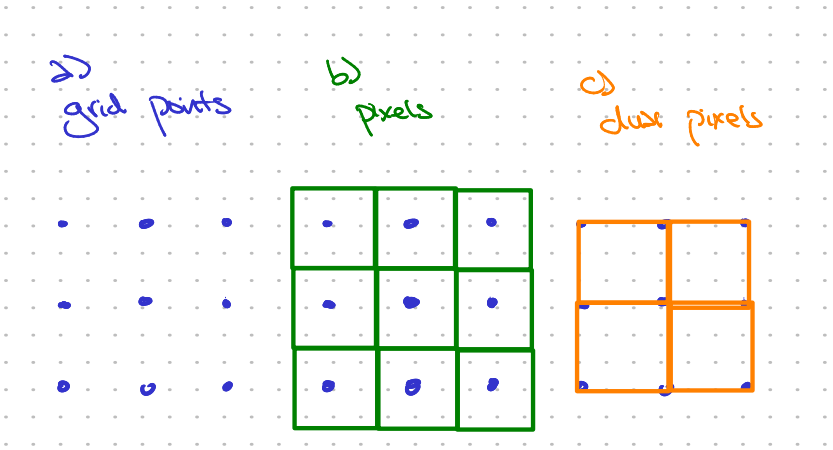


Figure 1: **Pixels and dual pixels:** pixels are unit squares centered at grid-points, with dual-pixels are unit-squares with corners at grid points.

1.1 feasibility

The question we need to ask is: how does our discretization relate to the continuous problem. There are two aspects to this: whether we can turn discrete solutions into continuous solutions, and how good it is.

The former is known as feasibility. If we let S include the pixels that overlap, then
Lemma: Any tour of S is a feasible tour for χ.

which follows from covering a superset of the region (i.e. the space of pixels in S) also covers R.

1.2 cost of the tour

Some terminology is apt. Let n be the number of vertices of R, S the set of centerpoints of the pixels that intersect R, and $N = |S|$.

To analyze this algorithm, which solves TSP on the discretized grid, we need to define what measures we care about. Indeed, there may be "hard" instances or "easy" instances. A natural objective is the *approximation ratio*, i.e. comparing the ratio of costs of this algorithm to the *optimal solution*, as discussed last week.

The optimal tour uses continuous points, so we cannot directly compare it to a tour of S. The first thing we need to do is to bring them onto the same axis:
Lemma: The optimal tour can be "snapped" onto the integers without increasing its length, such that all dual pixels that were previous touched are still touched.

where the relation between pixels, grid points, and dual pixels are shown in Fig. 1. While the discrete space looks the same, the continuous space differs, and we lose feasibility. The last important aspect to show is bound that we only lose points that are adjacent to the tour:
Lemma: A 2-wide square following the snapped tour covers S.

and therfore, we can walk around the boundary of the region swept by the 2-wide square, in additional linear cost (because the perimeter scales linearly). Intuitively, with a 2-wide square, it costs at most twice the length of the tour (indeed, this is true after a constant addition [1]).

Theorem 1.1. *If ℓ^* is the length of the optimal tour, then the length of TSP on grid cells is $\leq 3\ell^* + 6$.*

1.3 conclusion

We have summarized a case study of designing an approximation algorithm for a really quite difficult geometric problem. From turning it into a discrete problem (a graph on the pixels S that intersect R), and then proving multiple connections between the discrete and the continuous (snapping the grid only misses boundary points), we are able to derive an ≈ 3-approximation. We conclude with a few more comments on the specific result proved.

In this case, we do not lose anything if the grid is already integers and axis-aligned (as the loss was from the snapping); however solving TSP is still hard (later). Moreover, we assumed the cutter can only move orthogonally. Restricting arbitrary motion to orthogonal motion clearly costs a factor of at most $\beta \leq \sqrt{2}$[1].

Finally, solving the TSP is hard, so we can use approximation algorithms. For example, last week we talked about 2-approx for metric TSP; we will talk about 1.5-approx for metric TSP a bit later (giving us 6-approx and 4.5-approx respectively). There also exists a $(1+\epsilon)$ approx that runs in $N^{O(1/\epsilon)}$ time, giving us a poly-time $3+\epsilon$ approximation for lawn mowing. As a good balance, they also describe a $(53/40)$-approx algorithm for TSP in linear time (or $6/5$ if there are no holes), which we will discuss next week.

References

[1] Esther M. Arkin, Sándor P. Fekete, and Joseph S.B. Mitchell. "Approximation algorithms for lawn mowing and milling". In: *Computational Geometry* 17.1 (2000), pp. 25–50. ISSN: 0925-7721. DOI: https://doi.org/10.1016/S0925-7721(00)00015-8.

[1] a tighter bound of $\beta = 2/\sqrt{2+\sqrt{2}} \approx 1.08$ can be shown; see the paper